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SHORT BIO

I like building things end to end, covering product thinking and low-level implementation. I work across AI/ML and backend engineering, training models and shipping them to production with FastAPI, Docker, and cloud GPU providers. Following new research and tools helps me bring the right approach to the problem I am solving.

EXPERIENCE

AI/ML Engineer

Aug 2024 – Jul 2026

Coffee Inc.

Chennai, India

- Built AI tools for manufacturing clients, focusing on document intelligence and automated data extraction.
- Integrated open-source vision-language models, detection models, and specialized OCR models.
- Designed and containerized FastAPI microservices with Docker to serve ML inference endpoints in production.
- Deployed serverless GPU inference handlers on RunPod and Modal for VLM and LLM workloads to improve throughput and cost efficiency.

AI/ML Engineer Intern

Mar 2024 – Jul 2024

Coffee Inc.

Chennai, India

- Built SmartSearch to help a construction client search across their internal documents and the web. Also developed DocGPT to let users find answers directly from their uploaded documents.
- Prototyped Module Builder for developing agentic software and Mentor-Mentee to help course platforms onboard new users.
- Researched AI infrastructure by evaluating cloud GPU providers, vLLM inference, and model compression to estimate hosting costs and scalable deployments.
- Collaborated with the team to integrate AI into existing products.

ML Engineer Intern (Remote)

Mar 2023 – Feb 2024

PALgenie Technologies

Texas, US

- Assisted in evaluating LLM outputs by identifying data quality issues and documenting observations to support model improvements.
- Automated data preprocessing and validation tasks using Python to improve the efficiency and consistency of data preparation workflows.
- Documented data processing workflows and collaborated with engineers to ensure reproducible and scalable ML training pipelines.

EDUCATION

Velammal Engineering College

Chennai

Bachelor of Technology in Artificial Intelligence & Data Science

Aug. 2020 – Mar. 2024

PROJECTS

ADEOS | *Python, FastAPI, OCR, GPU Inference*

- Designed and built the backend system and AI integration for an engineering document intelligence platform used by manufacturing companies.
- Developed FastAPI services to process manufacturing documents, handling use cases like ballooning and BOM extraction.
- Set up GPU inference pipelines to run OCR and vision-language models on technical drawings.
- Containerized the services with Docker for consistent deployment.

PFMEA Gap-Analysis Tool | *Python, LLMs, Excel Automation*

- Built an LLM-based gap-analysis tool for an automotive supply chain to automate the review of Process Failure Mode and Effects Analysis documents.
- Developed a pipeline where suppliers upload PFMEAs in any format and the system standardizes them automatically.
- Processed each document against historical quality data to flag missing failure modes, effects, and S/O/D scores.
- Directed Supplier Quality Engineers to review only the flagged low-confidence records instead of manually reviewing every record

Flowtic | *Python, Agents*

- Built a multi-agent workflow library that uses direct syntax to configure message routing without a heavy orchestration layer.
- Implemented isolated session management so each agent keeps its own conversation history, including images.
- Created a tool system where developers define functions once and agents call them automatically.
- Integrated LiteLLM to support any model provider

chatft | *Python, Unsloth, Transformers*

- Allows users to train chatbots on their WhatsApp chat data.
- Supports training any LLM on the given data with a focus on the Qwen series.
- Provides customizable preprocessing so users can decide how their data should look, such as deeper messages with more turns or a larger total sample size.
- Integrated with Unsloth, W&B, and Hugging Face.

nape-0 | *Python, Pytorch, Transformers*

- Nape is a small language model series that achieves comparative performance with larger models.
- It is computationally easy to deploy.
- Trained on **4x A6000s** in 5 hours.
- Total downloads on Huggingface: **39,125**

RESEARCH

CFG | *Python, Pytorch*

Dec 2022

- Consecutive Frame Generation (CFG) is a model built to generate video from a prompt-conditional input image.
- Core architecture uses a Transformer encoder for text, VQVAE for images, and a Unet as the decoder.

ADAPTIVE LEARNING WITH INVERSE REINFORCE LEARNING | *Python, Pytorch*

Mar 2023

- Combined behavioral cloning with reinforcement learning to reduce cold-start time and improve training accuracy. Validated on Atari environments.
- Currently extending this work to personalize LLMs at inference time.

TECHNICAL SKILLS

Cloud & GPU servers: Runpod, Lambda labs, AWS, Modal

Research interests: Natural Language Processing, computer vision, Reinforcement learning

Developer Tools: Git, Docker, Coolify

Libraries & Framework: Pytorch, Transformers (trl, datasets, sft), Unsloth, Deepspeed, Pytorch-lightning, Langchain, OpenAI Agents, FastAPI, Flask

Languages: Python, C, Dart (Flutter)